Jadd Akkad

416-409-5814 | akkadjadd@gmail.com | linkedin.com/in/jadd-akkad | github.com/Jaddboi | jaddboi.com

Education

University of Toronto Toronto, ON
Bachelor of Science in Computer Science, Mathematics, and Statistics *Present*
Relevant Coursework: Data Structures & Algorithms, Systems Programming, Operating Systems, Databases, Software Design, Software Engineering, Information Security, Computer Organization, Computational Complexity, Linear Algebra II, Statistics II, Intro to AI

Professional Experience

Nirvo Jan. 2026 - Apr. 2026
Fullstack Developer Intern

- **Led development of the admin dashboard** for a **Next.js/React.js** application using **TypeScript**.
- Built **user and session management** using **FastAPI** across overview and administration views.
- Added **account and session lifecycle controls** for deactivation, reactivation, deletion, editing, and cancellation.
- Implemented **filtering and pagination** for account and session records.
- Refined the landing page and dashboard with **responsive typography and interaction polish**.

Software Engineering Projects (jaddboi.com)

Gambot - Marvel Rivals Discord Bot | *JavaScript, Node.js, discord.js, RivalsIsland API* Oct 2026 - Present

- Developed a **Discord bot** for the **358-member UofT Marvel Rivals server** with interactive slash commands.
- Integrated the **RivalsIsland API** for season/mode-filtered player statistics and recent match history.
- Built **paginated stat embeds and match scoreboards** with user-restricted navigation and lazy loading.
- Added **in-memory caching, rate-limit handling, command cooldowns,** and automated tests.

Connect Four AI Agent | *Python, Game Theory, Heuristics* Oct. 2026

- Built an adversarial agent using **Minimax with Alpha-Beta Pruning** for a 6x7 board.
- Designed a **heuristic evaluation function** that prioritized strategic positioning and won over **80% of simulated matches**.
- Added **dynamic move ordering** to increase pruning and reduce computation time.
- Tuned **recursion depth** to balance decision quality with real-time performance.

Hush - Full-Stack Real-Time Messaging Platform | *MERN Stack, Socket.IO, Docker* Jul. 2026

- Built **direct and group chat** with typing indicators, presence, and delivery/read receipts.
- Added **reactions, replies, search, pins, and rich-media sharing**.
- Enforced **backend authorization** for friendships, groups, administration, blocking, reporting, and moderation.
- Secured webhooks, validated server input, **rate-limited uploads**, and protected media access.
- Added **cursor pagination** and synchronized client state through **Socket.IO events**.
- Built **backend/frontend tests** and a production-ready multi-stage **Docker deployment**.

StudyFlow - AI Study Platform (In Progress) | *Next.js, Supabase, Gemini API* Jul. 2026 - Present

- Architected a **full-stack academic platform** for courses, assignments, calendars, and grades.
- Built **upload workflows** for course documents and learning materials.
- Added **PDF extraction, text chunking, vector embeddings, and semantic retrieval**.
- Generated **automated summaries and structured study notes** with the Gemini API.
- Secured data with **authentication, RLS, private storage, and ownership validation**.
- Added quotas, atomic job claiming, persistent states, validation, **retry/fallback handling**, and **51 automated tests**.

Coding Quest Competition | *Competitive Programming* Oct. 2022

- Represented the International School of Kuala Lumpur and placed **1st overall**.
- Earned multiple **top-10, top-20, and top-30 global problem solves**.

Technical Skills

Languages: Python, Java, C, SQL, JavaScript, TypeScript, HTML/CSS, Bash, R, RISC-V Assembly
Frameworks & Libraries: Next.js, React.js, Express.js, FastAPI, LangChain, LangGraph, Tailwind CSS, Socket.IO, discord.js
Databases & Backend: PostgreSQL, MongoDB, Mongoose, Supabase, REST APIs
Tools & Systems: Git/GitHub, Docker, Linux/Unix, Windows, macOS, Cursor, VS Code, IntelliJ
Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, MVC Architecture, Unit Testing, Agile Development, Prompt Engineering